Corporate Finance

Lecture 8 extras: How a project is evaluated

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How a project is evaluated

- A project costs I today and pays a cash flow c at the end of each of the next N years. Its **cost of capital** r is the return investors could earn elsewhere at the same risk (Lecture 6).
- **Way 1: compute a return, then compare it with the cost of capital.** Ask what return the project itself earns: the rate that makes the NPV zero. That rate is the **internal rate of return** (IRR) :

$$\text{NPV} = -I + \frac{c}{1+\text{IRR}} + \frac{c}{(1+\text{IRR})^2} + \cdots + \frac{c}{(1+\text{IRR})^N} = 0$$

$$I = \frac{c}{1+\text{IRR}} + \frac{c}{(1+\text{IRR})^2} + \cdots + \frac{c}{(1+\text{IRR})^N}$$

- Solve for IRR. Accept the project when the IRR is above the cost of capital: the project earns more than the money costs. Here the test is a rate against a rate.

How a project is evaluated (con't)

- **Way 2: discount, then compare amounts.** Do not compute the project's own return. Instead, discount every c at the cost of capital r and add up:

$$\text{PV} = \frac{c}{1+r} + \frac{c}{(1+r)^2} + \cdots + \frac{c}{(1+r)^N} \qquad \text{NPV} = \text{PV} - I$$

- Accept when the NPV is positive: the project is worth more than it costs.
- Here the **discount rate** is the cost of capital. Once it has been used, there is nothing left to compare it with: the test is an amount against an amount, PV against I.

Two equivalent approaches: a return above the cost of capital is exactly what makes the discounted cash flows exceed the investment. The two analyses give the same answer.